

Clinical Dermatology

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Office of Academic Affairs

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Session Objectives

- Review aspects of the HPI that are critical for assessing a dermatological complaint.
- Recognize primary vs. secondary and shapes, patterns and configurations of skin lesions.
- Review definitions of exam findings of the hair and the abnormal patterns of hair loss and hair growth.
- Review nail anatomy and common abnormal findings.
- Understand risk stratification for melanoma.

Disclaimers

- Reaching a final diagnosis for dermatological complaints can be difficult, but if you know how to properly use terminology to describe lesions you will be more successful in identifying differential diagnoses to evaluate further
- I have included photos of skin lesions across a range of skin tones. However, to keep this presentation manageable, I couldn't provide an image of each lesion on every skin color. For additional examples of how skin lesions present on various skin tones, I encourage you to utilize VisualDx and other resources provided in the CPG.

Subjective

- Prior to walking in the room, always consider what you already know
 - Patient demographics: age, gender, ethnicity
 - Vital signs: especially if this is an acute presentation
 - Chief complaints
 - Is it a rash?
 - Is it a single skin lesion? Multiple skin lesions?
 - Is it a lump?
 - Is it hair loss? Hair growth?

HPI

- Acute vs. chronic
 - If it's chronic how has it changed over time
- Onset: gradual vs. sudden
- Location, location, location
 - Generalized or diffuse
 - Unilateral vs. bilateral
 - Sun exposed skin: head, ears, face, neck, forearms, hands
 - Truncal: chest, back, abdomen

HPI

- Location, location, location continued
 - Extremities
 - Acral: affecting distal portions of limbs
 - Extensor surfaces: posterior arm and forearm, anterior thigh and leg
 - Flexor or Flexural surfaces: antecubital fossa, popliteal fossa
 - Palms and soles
 - Intertriginous: areas where two skin surfaces contact each other
 - Axilla, inguinal, intergluteal, inframammary
 - Webspaces between fingers and toes
 - Are mucosal surfaces affected?
 - Does the rash involve the scalp or other areas of hair growth?

HPI

- Progression: how has it changed over time?
 - Constant vs. intermittent
 - If intermittent, does it occur at a certain time of the day or in association with a specific activity?
 - Did it start in one area and spread to other areas?
 - Centrifugal spread: rash that started on the trunk and spread to the extremities
 - Centripetal spread: rash that started on the extremities and spread to the trunk
 - Cephalic to caudal spread: rash that started on the head and spread down towards the feet

HPI

- Preceding triggers or exposures
 - Recent travel: Travel to endemic areas?
 - Recent outdoor excursion:
 - Hiking in wooded areas?
 - Animal exposure?
 - Insect bites?
 - Recent sun exposure?
 - Close contact
 - Does anyone you share a living space with have a similar rash?

HPI

- Preceding triggers and exposures continued
 - New medications? New supplements?
 - New laundry detergent? New clothing?
 - New shampoo, lotion, body wash, face wash or make-up?
 - Occupational exposures:
 - Contact with chemicals?
 - Airborne exposures?

Associated Symptoms

- Associated symptoms are different than review of systems
- Associated symptoms are signs and symptoms that help you differentiate between the most likely diagnoses for the chief complaint
- Associated symptoms are signs and symptoms that can also indicate severity or progression of the diagnoses for the chief complaint
- Is the rash itchy? Painful? Numb?
- Have they experienced fever? Fatigue? Joint pain?

Past Medical History

- How to acquire a past medical history has already been taught to you
- The lecture, Dermatologic Signs of Systemic Disease, will expand on when past medical history can be relevant to a dermatologic chief complaint

Exam

- How to describe what you see and feel
- Palpation is necessary to fully describe what a rash or lesion is

Primary and Secondary Lesions

- Definitions
 - Primary: physical changes in the skin caused directly by the disease process
 - Secondary: may evolve from primary lesions or may be caused by external forces such as scratching, trauma, infection or healing

Primary Skin Lesions

- Macule: circumscribed area <1 cm that is flat and level with skin surface differing only in color from surrounding skin
- Patch: circumscribed area >1 cm that is flat and level with skin surface differing only in color from surrounding skin
- Papule: solid, circumscribed lesion <1 cm that is elevated above the skin surface
- Nodule: solid, circumscribed lesion 1-5 cm in size that is elevated above the skin surface
- Tumor: solid, circumscribed mass >5 cm that is elevated above the skin surface

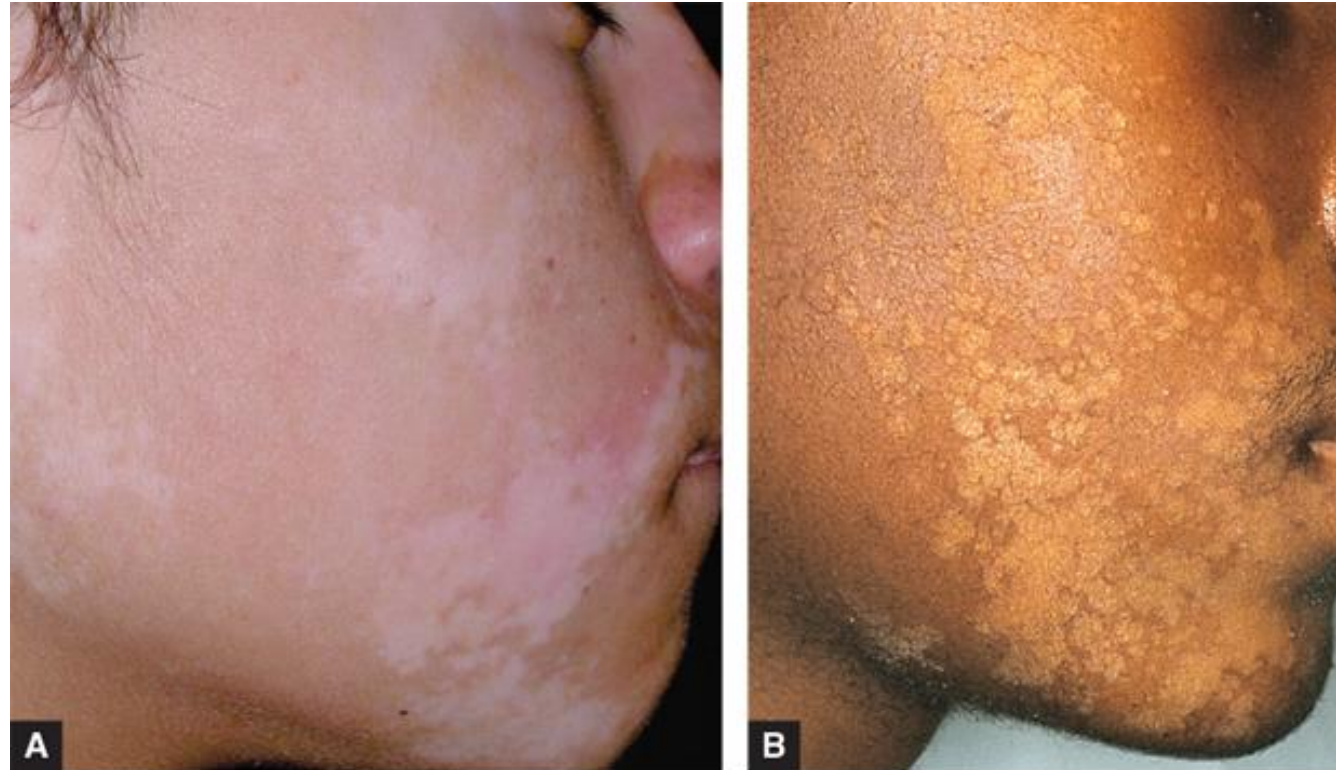
Primary Skin Lesions

- Plaque: solid, circumscribed lesion >1 cm that is broadly elevated
- Vesicle: elevated, circumscribed serous fluid filled lesion <1 cm
- Bullae: elevated, circumscribed serous fluid filled lesion >1 cm
- Pustule: elevated, circumscribed purulent fluid filled lesion
- Wheal: elevated, erythematous area of dermal edema (aka welts, hives or urticaria)



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Many tan macules (solar lentigines) and a black macule with a spiculated border (ink spot lentigo) on the posterior neck.



Source: Susan Burgin
Guidebook to Dermatologic Diagnosis
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(A) Depigmented macules of vitiligo (Used with permission from Dr. Miguel Sanchez, New York University, New York, NY.); (B) hypopigmented macules of tinea versicolor.



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Well-demarcated oval, brown patches at the preauricular area.



Source: Susan Burgin
Guidebook to Dermatologic Diagnosis
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Dome-shaped umbilicated papules of molluscum contagiosum. (Reproduced with permission from Kang S, Amagai M, Bruckner AL, et al: Fitzpatrick's Dermatology, 9th ed. New York, NY: McGraw Hill; 2019.)



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Several dark brown, stuck-on, verrucous papules on the temple and cheek, and around the eye.



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Nodule, lymphoma cutis.



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A pink and dark brown nodule with scant secondary scale on the leg.



A close-up of a thin, violaceous, crusted plaque on the forehead (superficial pigmented BCC).



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A close-up of an actinic keratosis showing a thin, pink, gritty papule against a background of dermatoheliosis.



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A close-up of a cluster of tiny vesicles.



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Vesicles and bullae, linear IgA disease.



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Bullous arthropod bites showing many tense and flaccid vesicles and bullae on the extensor arm.



ed with infectious folliculitis due to nonbacterial causes
A large cluster of follicularly based pustules and scattered papules on the arm.



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but have been associated with a pruritic periorificial papulopustular eruption in
infant adults or with a more widespread eruption. Numerous large pustules on an extremity.



Source: Susan Burgin
Guidebook to Dermatologic Diagnosis
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Erythema toxicum neonatorum, showing tiny pustules on the leg.



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine, 8th Edition*: www.accessmedicine.com

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Widespread large, pink, edematous welts on the arm and trunk.

Secondary Skin Lesions

- Scale: accumulation of loose or adherent desquamated layers of skin (typically keratinized cells of stratum corneum)
 - Xerosis is the term for dry skin
- Crust: dried remains of fluid overlying areas of damaged skin (can be blood, purulent fluid, serous fluid)
- Eschar: type of crust indicating skin necrosis
- Erosion: depression in the surface of the skin from loss of superficial epidermis
- Excoriation: a type of erosion caused by abrasion or scratching
- Ulceration: depression in the surface of the skin from loss of the epidermis and some or all of the dermis

Secondary Skin Lesions

- Atrophy: thinning of skin layers
 - Epidermis thinning results in fine wrinkling
 - Dermal thinning results in a depression in the surface of the skin
- Fissure: linear cleavage or crack in the skin extending into the dermis
- Lichenification: thickening of the epidermis with accentuation of skin markings
- Scar: area of fibrous tissue proliferation that replaces area of skin that suffered damage to the dermis
 - Hypertrophic: scar that is elevated but confined to original borders
 - Keloid: excessive fibrous tissue proliferation leading to a scar that is elevated and spread beyond the original borders



Source: Susan Burgin
Guidebook to Dermatologic Diagnosis
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A scaly plaque of psoriasis.



A close-up of a violaceous and whitish plaque with some overlying scale.



An eroded plaque with purulent crust (secondary infection) around the eye.



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Primary HSV showing many erosions scattered over the vulva.



A curvilinear, crusted erosion and a nearby similar small crust on the forehead.



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine, 8th Edition*: www.accessmedicine.com

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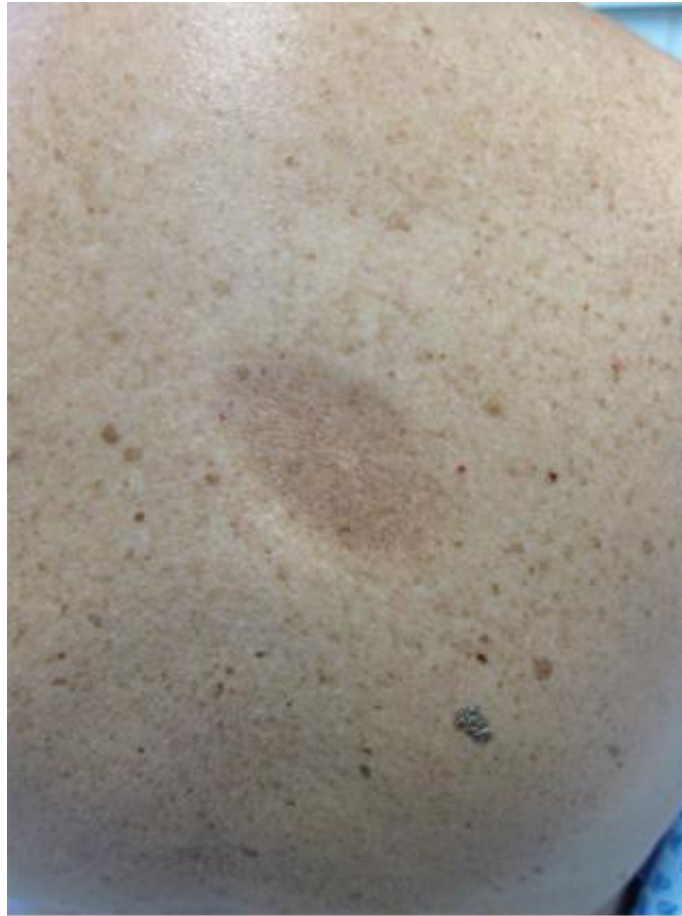


Ulcerated plaques with overlying hemorrhagic crust on the cheek.



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Eschars overlying ulcers and associated edema of the penis.



Source: Susan Burgin
Guidebook to Dermatologic Diagnosis
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Atrophoderma of Pasini and Pierini with light brown depressed plaques with “cliff-drop borders.”





Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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Hyperkeratotic plaques with associated fissures on the palm.



Source: Goldsmith LA, Katz SI, Gilchrest BA, Paller AS, Leffell DJ, Wolff K: *Fitzpatrick's Dermatology in General Medicine, 8th Edition*: www.accessmedicine.com

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A close-up of a lichenified scaly plaque with follicular prominence.



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A lichenified, hyperpigmented, scaly plaque on the dorsal hand near the webspace.



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A close-up of a hypertrophic scar, appearing as a linear shiny plaque.



A close-up of two linear, hyperpigmented keloidal plaques.

Shape and Configuration

- Shape describes an individual primary lesion
- Configuration describes an arrangement of individual primary lesions

Shape

- Annular: ring shaped with variation in appearance between periphery and center
- Discoid or nummular: circular or coin shaped with uniform appearance
- Ovoid: round with elliptical or oval shape with uniform appearance
- Targetoid: ring shaped with two distinct zones that vary in appearance
- Arcuate: incomplete ring or arc shaped
- Stellate: having multiple angulated edges, resembling a star
- Linear



edema of the lips, eyelids, hands, and feet. It may also affect the cheeks, tongue,

and genitalia. Edematous and erythematous annular plaques on the trunk of a child with drug-induced urticaria.



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occurs on the lips, eyelids, palms, and feet may also affect the cheeks, tongue, and genital. Arcuate, annular, and polycyclic pink plaques with lighter centers on the chest of a child with urticaria multiforme.



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Nummular lesion, nummular dermatitis.



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Atypical targetoid lesions, Stevens-Johnson syndrome due to medication.



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Classic target lesions (with 3 zones of color) on the palm.

Configuration

- Annular: circular arrangement of lesions
- Linear: arrangement of lesions forming a line or band
- Reticulate: arrangement of lesions that has a net-like or lacy appearance
- Clustered: multiple lesions adjacent to each other
- Confluent: multiple lesions joining together
- Discrete: distinct lesions, separate from each other
- Geometric: sharp borders or with a pattern that suggests the lesion was derived by an object against the skin



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Superficial erosions, some in a linear array, on the cheek.



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is peripheral motor neuropathies, neurogenic bladder, and diaphragmatic
related with zoster. Both herpes zoster and herpes simplex virus can cause

A close-up of tiny grouped and confluent vesicles on a thin erythematous plaque.



Grouped vesicles becoming confluent to form bullae with outlying erythema and scattered pink papules.



Reticulate red-brown erythema on the legs.



An erythematous plaque with a geometric shape on the forearm secondary to application of a topical steroid-containing tape.

Pattern

- Dermatomal: lesions following a dermatome of a single spinal afferent nerve root, unilateral, does not cross midline
- Guttate: lesions look as if they were dropped on the skin using a dropper
- Serpiginous: lesions wander as if following the track of a snake
- Follicular: lesions centered around hair follicles



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Grouped vesicles on an erythematous base in a T3 distribution and few scattered papules outside the dermatome.



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Several discrete, well-demarcated, light pink, scaly, small plaques on the trunk.



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Serpiginous erythema, jellyfish sting.



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A close-up of multiple follicularly-based, scaly papules.



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A close-up of follicularly based pustules, each with surrounding erythema. Note the central hair in each.



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Scaly and smooth papules in the webspaces and on the fingers.

Hair

- Hair loss
 - Effluvium or defluvium: shedding of hair
 - Baldness is the resulting condition called alopecia
 - Alopecia can be cicatricial (scarring) or noncicatricial (non scarring)
 - Absence of follicular orifices distinguishes between them
- Excessive hair growth
 - Hirsutism: excess hair growth in women in a male distribution pattern
 - Hypertrichosis: excess hair growth for age, race and sex in non male distribution pattern
 - Hair density or length beyond accepted limits of normal



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Terminal dark brown hairs on the upper and lower cutaneous lip.



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A close-up of a large brown plaque with associated hypertrichosis.



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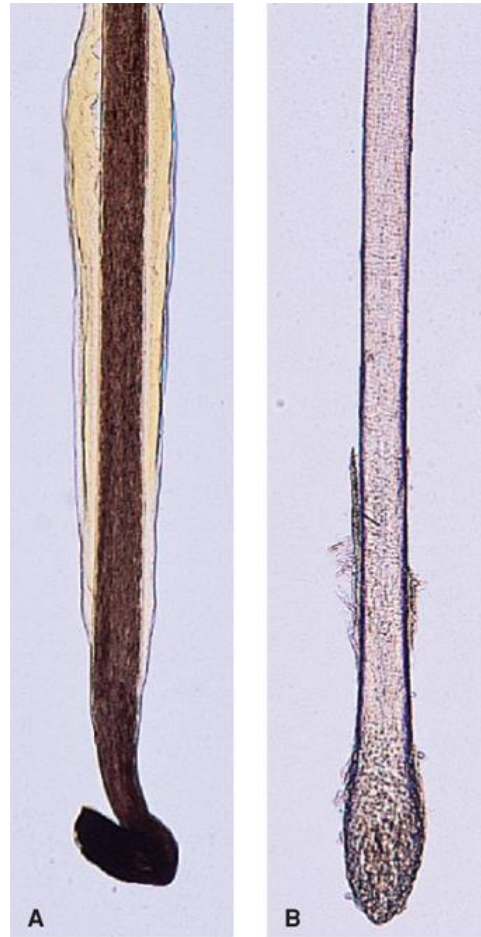
Terminal hair growth on the central chest and around the areolae.

Exam

- Is the hair fine or coarse?
 - Is each hair the same diameter/caliber?
- Are there areas of thinning or decreased density?
 - Assess density by amount of space between parts
- Are there short, broken hairs that are easy to extract?
 - These are called exclamation point hairs

Exam

- Hair pull test
 - Grab 50-60 hairs at the base and pull firmly
 - Extraction of more than 6 hairs is positive for active hair loss
 - Are the extracted hairs anagen or telogen hairs?
- Trichogram
 - Plucking approximately 50 hairs and counting number of anagen and telogen hairs
- Trichoscopy (using a dermatoscope)



Source: Arturo P. Saavedra, Ellen K. Roh, Anar Mikailov
Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, 9e
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Hair mount (A) Anagen: Note the malleable proximal ends and (B) Telogen: club hairs. (Reproduced with permission from Goldsmith LA, Katz SI, Gilchrist BA, et al., eds. Fitzpatrick's Dermatology in General Medicine. 8th ed. New York, NY: McGraw-Hill; 2012.)



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A close-up of nonscarring hair loss and some exclamation point hairs.



Lichen planopilaris

Dermoscopic image of lichen planopilaris presenting with irregularly distributed follicular openings (red arrows) denoting scarring (cicatricial)

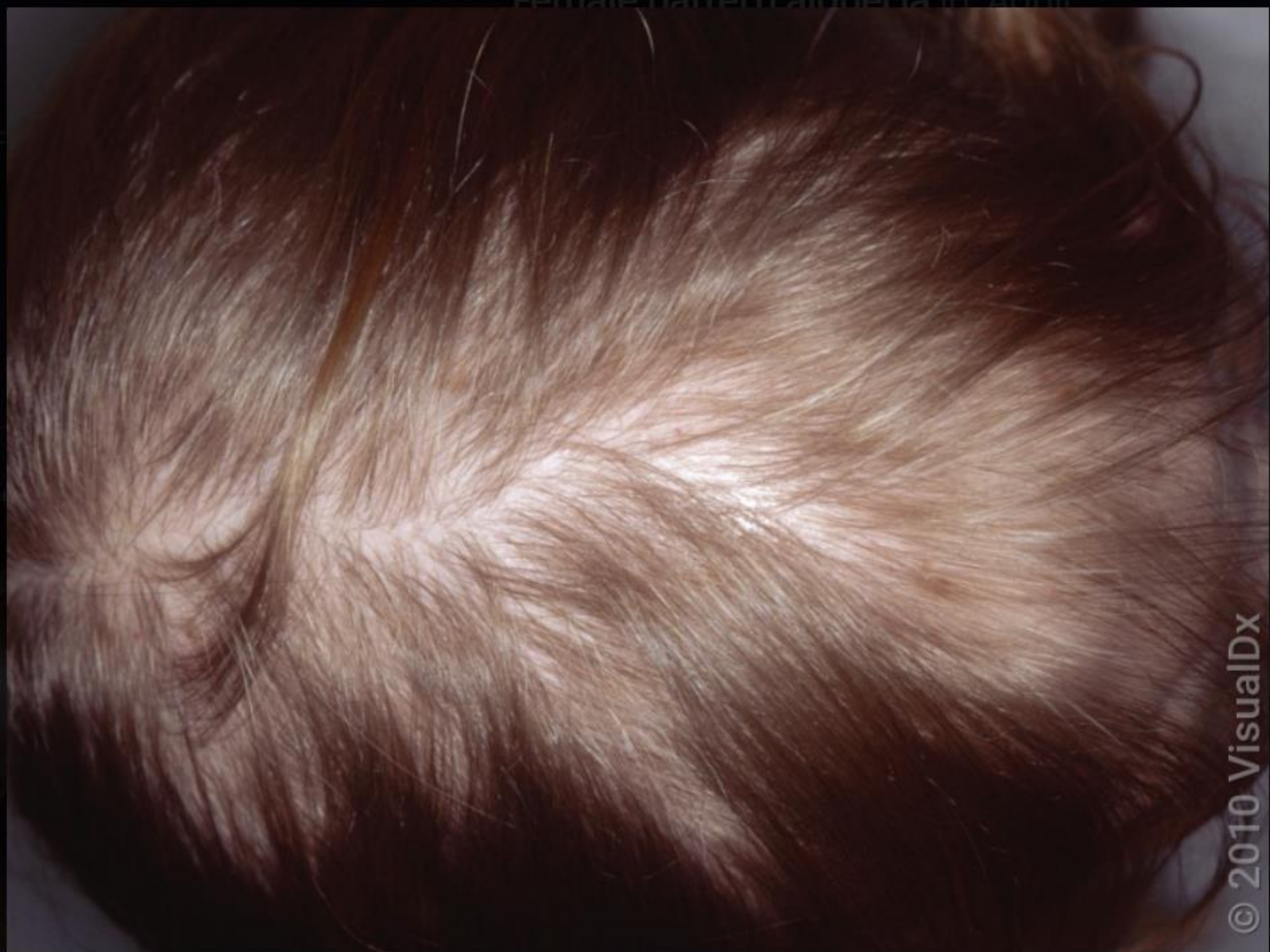
Source: dermnetnz.org

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<https://dermnetnz.org/topics/trichoscopy>

Common Conditions

- Pattern hair loss or androgenic alopecia
 - Most common type of progressive hair loss
- Alopecia areata: localized loss of hair in round or oval areas
- Telogen effluvium: transient increased shedding of normal hairs
 - Patients most commonly present with complaint of increased daily hair loss with brushing
- Traction alopecia can be transient but if prolonged can lead to a cicatricial alopecia



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Diffuse mild thinning of the hair on the crown and widening of the midline part.



Diffuse thinning of the hair on the crown and widening of the midline part.



A smooth round patch of nonscarring alopecia on the occipital scalp with early hair regrowth after intralesional steroid injection.



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Round patches of nonscarring hair loss in the beard area.



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Decreased hair density as seen at the anterior scalp.



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Decreased hair density as seen on the vertex.

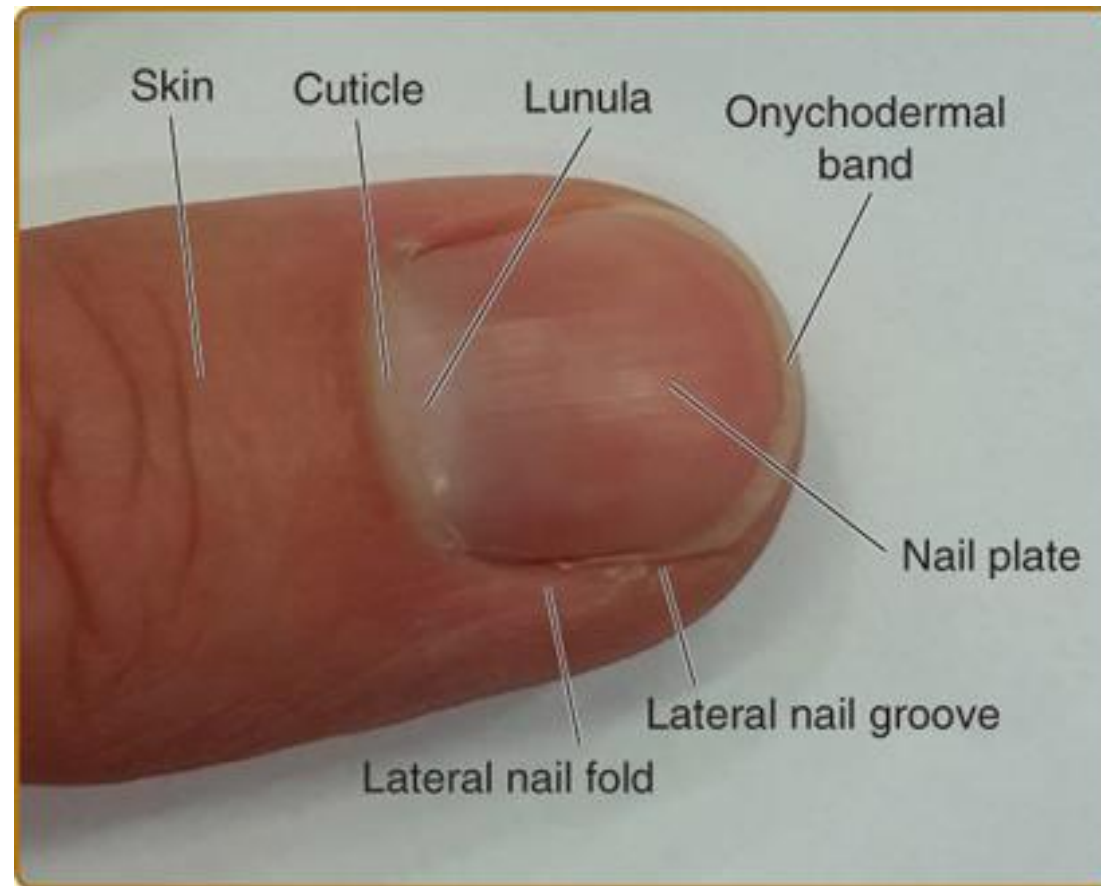


Patchy non-scarring alopecia in a band across the anterolateral hairline. Note also the presence of the fringe sign.



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A pink and whitish plaque with focal scarring alopecia on the scalp.



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Picture of a human fingertip with the nail organ. Visible structure include the skin, cuticle, nail plate with the proximally located lunula, and the distally located onychodermal band. On the side, the nail plate is separated by the lateral nail groove and lateral nail fold.

Nails

- Ask patients if they have noticed changes to
 - Nail color
 - Nail surface texture
 - Nail thickness
 - Underneath the nail (subungual)
- Chromonychia: nail discoloration
 - Leukonychia is the most common (white discoloration)
- Surface texture abnormalities include
 - Brittle nails, longitudinal and transverse grooves, pitting

Nail Surface Texture

- Brittle nails
 - Onychorrehexis: distal longitudinal splitting
 - Onychoschizia: horizontal or lamellar splitting at nail's free end
- Nail pitting: small depressions in the nail surface resulting from abnormal keratinization
- Trachyonychia: rough nails
- Onycholysis: detachment of the nail from the distal nail bed



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: *Fitzpatrick's Dermatology*, Ninth Edition
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Onychoschizia.



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Coarse pits of the nails and some distal onycholysis.



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Trachyonychia in a patient with alopecia areata.



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Roughened discolored nail plates.

Melanoma

HARMM Risk factors

- **H**istory of previous melanoma
- **A**ge over 50
- **R**egular dermatologist absent
- **M**ole changing
- **M**ale gender

Other risk factors:

- 50 or more common moles
- Red/light hair color/light eye color
- Freckles
- Increased sun exposure (including UV radiation from tanning booths)
- Immunosuppression
- Family history

Melanoma

- **A**symmetry of one side of the mole compared to the other
- **B**orders irregular; ragged, notched, blurred
- **C**olor change, especially blue or black
- **D**iameter $\geq 6\text{mm}$
- **E**volution or change in size, symptoms and morphology

THE ABCDEs OF MELANOMA

What to Look for:

Melanoma is the deadliest form of skin cancer. However, when detected early, melanoma can be effectively treated. You can identify the warning signs of melanoma by looking for the following:



One half is unlike the other half.



Irregular, scalloped or poorly defined border.



Varied from one area to another; shades of tan and brown, black; sometimes white, red or blue.



While melanomas are usually greater than 6mm (the size of a pencil eraser) when diagnosed, they can be smaller.



A mole or skin lesion that looks different from the rest or is changing in size, shape or color.

Example:



References

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